3

Signal Processing

3.1 Introduction

Sequential signals are surprisingly widespread in the laboratory and require a large number of methods for analysis. Most data are obtained via computerised instruments such as near infrared (NIR), high performance liquid chromatography (HPLC) or nuclear magnetic resonance (NMR), and the nature of information such as peak integrals, shifts and positions is often dependent on how the information from the computer is first processed. An appreciation of this step is often essential before applying further multivariate methods such as pattern recognition or classification. Spectra and chromatograms are examples of series that are sequential in time or frequency. However, time series also occur very widely, for example, in industrial process control and natural processes.

3.1.1 Environmental and Geological Processes

An important source of data involves recording of samples regularly with time. Classically, such time series occur in environmental chemistry and geochemistry. A river might be sampled for the presence of pollutants such as polyaromatic hydrocarbons or heavy metals at different times of the year. Is there a trend and can this be related to seasonal factors? Different and fascinating processes occur in rocks, where depth in the sediment relates to burial time. For example, isotope ratios are a function of climate, as relative evaporation rates of different isotopes are dependent on temperature: certain specific cyclical changes in the Earth's rotation have resulted in the ice ages and so climate changes, leaving a systematic chemical record. A whole series of methods for time series analysis primarily based on the idea of correlograms (Section 3.4) can be applied to explore such types of cyclicity, which are often quite hard to elucidate. Many of these approaches were first used by economists and geologists who also encountered related problems.

One of the difficulties is that long-term and interesting trends are often buried within short-term random fluctuations. Statisticians distinguish among various types of noise that interfere with the signal, as discussed in Section 3.2.3.

In addition to obtaining correlograms, a large battery of methods are available to smooth time series, many of which are based on the so-called 'windows', whereby data are smoothed over a number of points in time. A simple method is to take the average reading over five points in time, but sometimes, this could miss out important information about cyclicity, especially for a process that is sampled slowly compared with the rate of oscillation. A number of linear filters have been developed, which are applicable to this type of data (Section 3.3); this procedure is often being described as convolution.

3.1.2 Industrial Process Control

In industry, sequential series may occur in the manufacturing process of a product. It could be crucial that a drug has a certain well-defined composition; otherwise an entire batch is unmarketable. Sampling the product regularly in time is essential for two reasons. The first is monitoring, simply to determine whether the quality is within the acceptable limits. The second is for control, to predict the future and check whether the process is getting out of control. It is costly to destroy a batch and not economically satisfactory to obtain information about acceptability several days after the event. As soon as the process begins to go wrong, it is often advisable to stop the plant and investigate. However, too many false alarms can be equally inefficient. A whole series of methods have been developed for the control of manufacturing processes, an area where chemometrics can often play a key and crucial role. In this chapter, we will not be discussing statistical control charts in detail, the whole topic being worthy of a book in its own rights. However, a number of methods outlined in this chapter are useful for handling such sequential processes,

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especially to determine whether there are long-term trends that are gradually influencing the composition or nature of a manufactured product. Several linear filters together with modifications such as running median smoothing (RMS) (Section 3.3) can be employed under such circumstances. Chemometricians are especially interested in the extension to multivariate methods, for example, monitoring a spectrum as recorded regularly in time, which will be outlined in detail in later chapters.

3.1.3 Chromatograms and Spectra

The most common applications of methods for handling sequential series in the laboratory arise in chromatography and spectroscopy and will be emphasised in this chapter. An important aim is to smooth a chromatogram. A number of methods such as the Savitzky–Golay filter have been developed here (Section 3.3.1.2). A problem is that if a chromatogram is smoothed too much, the peaks are blurred and lose resolution, negating the benefits; thus, optimal filters have been developed that remove noise without broadening peaks excessively.

Another common need is to increase resolution, and sometimes spectra are routinely displayed in the derivative mode (e.g. electron spin resonance spectroscopy): there are a number of rapid computational methods for such calculations that do not emphasise noise too much (Section 3.3.2). Other approaches based on curve fitting and Fourier filters are also very common.

3.1.4 Fourier Transforms

The Fourier transform (FT) has revolutionised spectroscopy such as NMR and IR since the 1950s. The raw data are not obtained as a comprehensible spectrum but as a time series, where all spectroscopic information is mixed up and a mathematical transformation is required to obtain a comprehensible spectrum. One reason for performing FT spectroscopy is that a spectrum of acceptable signal-to-noise ratio can be recorded much more rapidly than via conventional spectrometers, often a hundred times more rapidly. This has allowed the development, for example, of ^{13}C NMR, as a routine analytical tool, because the low abundance of ^{13}C is compensated by faster data acquisition. However, special methods are required to convert this 'time domain' information (called a *free induction decay* (FID) in NMR parlance) into a 'frequency domain' spectrum, which can be interpreted directly (see Section 3.5.1).

Parallel with Fourier transform, spectroscopy has arisen a large number of approaches for enhancement of the quality of such data, often called *Fourier deconvolution*, involving manipulating the time series before Fourier transformation (Section 3.5.2). Many of these filters have their origins in engineering and are often described as digital filters. These are quite different to the classical methods for time series analysis used in economics or geology. Sometimes, it is even possible to take non-Fourier data, such as a normal spectrum, and Fourier transform it back to a time series, then use Fourier deconvolution and Fourier transform back again, often called *Fourier self-deconvolution*.

Fourier filters can be related to linear methods discussed in Section 3.3 by the convolution theorem (as discussed in Section 3.5.3).

3.1.5 Advanced Methods

In data analysis, there will always be new computational approaches that promote great interest among statisticians and computer scientists. Much frontline research in chemometrics is involved refining such methods, but it takes several years before the practical worth or otherwise novel data analytical approaches is demonstrated.

Methods for so-called *non-linear deconvolution* have been developed over the past few years, one of the most well-known methods being maximum entropy (Section 3.6.4). This latter approach was first used in infrared astronomy to deblur weak images of the sky and been successfully applied, for example, to police photography to determine car number plates from poor photographic images of a moving car in a crime. Enhancing the quality of a spectrum can also be regarded as a form of image enhancement and thus use similar computational approaches. A very successful application is in NMR imaging for medical tomography. The methods are called *non-linear* because they do not insist that the improved image is a linear function of the original data. A number of other approaches are also available in the literature, but maximum entropy has received much publicity largely because of the readily available software.

Wavelet transforms (Section 3.6.2) involve fitting a spectrum or chromatogram to a series of functions based on a basic shape called a *wavelet*, of which there are several in the literature. These transforms have the advantage in that, instead of storing, for example, 1024 spectral data points, it may be possible to retain only a few most significant wavelets and still not lose much information. This can result in both data compression and denoising of data.

Rapid algorithms for real-time filtering have attracted much interest among engineers and can be used to follow a process by smoothing the data as it occur. The Kalman filter is one such method (Section 3.6.1) that has been reported extensively in the analytical chemistry literature. Early textbooks of chemometrics often featured Kalman filters; although they still remain a tool, they are less prominent in the chemometrics literature nowadays but are widely used by engineers.

3.2 Basics

3.2.1 Peak shapes

Chromatograms and spectra are normally considered to consist of a series of peaks, superimposed upon noise. Each peak usually arises either from a characteristic absorption (such as a chromophore) or a characteristic compound. In most cases, the underlying peaks are distorted for a variety of reasons such as noise, blurring or overlap with neighbouring peaks. A major aim of chemometric methods is to obtain the underlying, undistorted information.

Peaks can be characterised in a number of ways, but a common approach, for symmetrical peaks as illustrated in Figure 3.1, is to characterise each peak by

1) a position at the centre (e.g. the elution time or spectral frequency),
2) a width, normally at half height and
3) an area.

For symmetrical peaks, the position in the centre is also normally the position of maximum height (we will define peaks as being unimodal) and the mean. Peak shapes can usually be related to statistical distributions, and in quantum mechanical terms are in fact such.

The relationship between area and peak height is dependent on peak shape, as discussed below, although heights are often easier to measure. If a series of peaks have the same shape, then the ratios of heights are proportional to ratios of areas. However, area is usually a better measure of chemical properties such as concentration and it is important to obtain precise information about peak shapes before relying on heights, for example, as raw data for pattern recognition programs.

Sometimes, the width at a different percentage of the peak height is cited rather than the half width. A further common measure is when the peak has decayed to a small percentage of the overall height (e.g. 1%), which is often taken as the total width of the peak, or alternatively has decayed to a size that relates to the noise level, for example, the root mean square noise.

In many cases of spectroscopy, peak shapes can be very precisely predicted, for example, from quantum mechanics, such as in NMR or visible spectroscopy. In other situations, the peak shape is dependent on complex physical processes, for example, in chromatography, and can only be modelled empirically. In the latter situation, it is not always practicable to obtain an exact model, and a number of closely similar empirical estimates will give equally useful information.

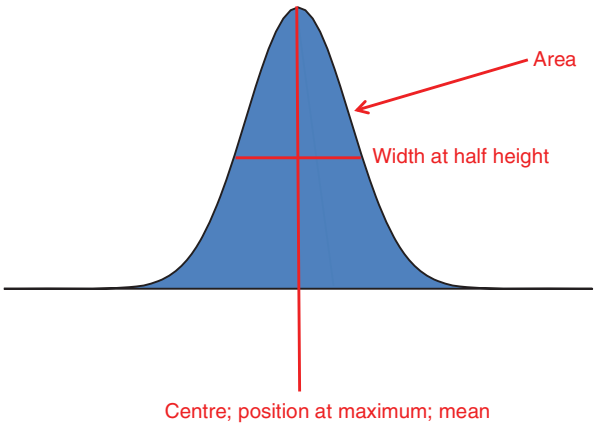


Figure 3.1 Main parameters that characterise a symmetric peak.

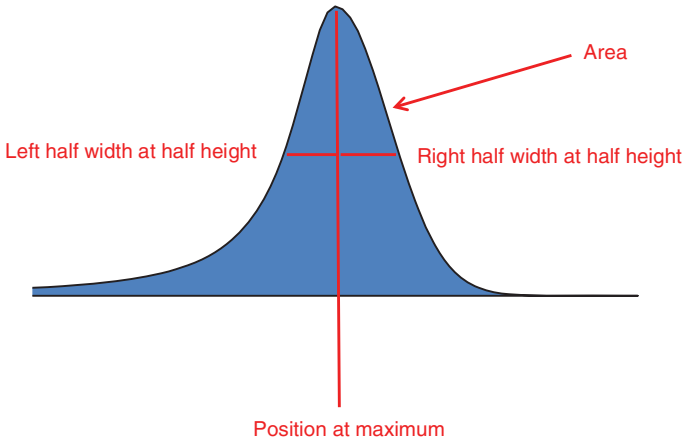


Figure 3.2 Main parameters that characterise an asymmetric peak.

For asymmetric peak shapes, it is normal to use the position maximum rather than the mean and to quote the left and right half widths at half height, as illustrated in Figure 3.2.

A few common peak shapes cover most situations. If these general peak shapes are not suitable for a particular purpose, it is probably best to consult specialised literature on the particular measurement technique. Note that unless there is a good reason, for example, from knowledge of quantum mechanics or a specific type of spectroscopy, it is usual to empirically model peak shapes with basic functions.

1. The relative peak heights of resolved compounds in a chromatogram are always proportional to relative concentrations.
   (a) True
   (b) False

3.2.1.1 Gaussians

These peak shapes are common in most types of chromatography and spectroscopy. A simplified formula for a Gaussian is given by

$$x_i = A \exp(-(x_i - x_0)^2 / s^2)$$

where

- A is the height at the centre,
- x_0 is the position of the centre and
- s relates to the peak width.

Gaussians are based on a normal distribution where

- x_0 corresponds to the mean of a series of measurements and
- $s/\sqrt{2}$ to the standard deviation.

It can be shown that

- the width at half height of a Gaussian peak is given by $\Delta_{1/2} = 2s\sqrt{\ln 2}$ and
- the area by $\sqrt{\pi}A\ s$ using the equation presented above: note that this depends on *both* the height and the width. Note that the height and area are in the original units; hence, if the width is measured in seconds and height in absorbance unit (AU), it will be in units of AU * s.

Notice that Gaussians are also the statistical basis of the normal distribution, see Section A.3.2, but the equation for the standard normal distribution is normally scaled so that the area under the curve equals 1. For signal analysis, we will use the simplified expression above.

1. A Gaussian peak has a height of 5 units and a width at half height of 2 units. Its area in the original units^2 is
   (a) 10
   (b) 10.64
   (c) 16.15
2. A peak has an area of 25 units^2; the intensity is measured in AUs and the horizontal axis in minutes. What is its area in AU * s?
   (a) 0.416
   (b) 25
   (c) 1500

3.2.1.2 Lorentzians

The Lorentzian peak shape corresponds to a statistical function called the *Cauchy distribution*. It is less common but often arises in certain types of spectroscopy such as NMR. A simplified formula for a Lorentzian is given by

$$x_i = A/(1 + (x_i - x_0)^2/s^2)$$

where

- A is the height at the centre,
- x_0 is the position of the centre and
- s relates to the peak width.

It can be shown that

- the width at half height of a Lorentzian peak is given by $\Delta_{1/2} = 2s$ and
- the area by $\pi\ A\ s$: note that this depends on *both* the height and the width as per Gaussian.

The main difference between Gaussian and Lorentzian peak shapes is that the latter has a bigger tail, as illustrated in Figure 3.3, for two peaks with identical half widths and heights.

1. A Lorentzian peak has a height of 5 units and a width at half height of 2 units. Its area in the original units^2 is
   (a) 8.86
   (b) 10
   (c) 15.71
2. Lorentzian peaks have more pronounced tails than Gaussian peaks.
   (a) True
   (b) False

3.2.1.3 Asymmetric Peak Shapes

In many forms of chromatography, it is hard to obtain symmetric peak shapes. Although a number of quite sophisticated models are available, a very simple first approximation is that of a Gaussian/Lorentzian peak shape. Figure 3.4(a)

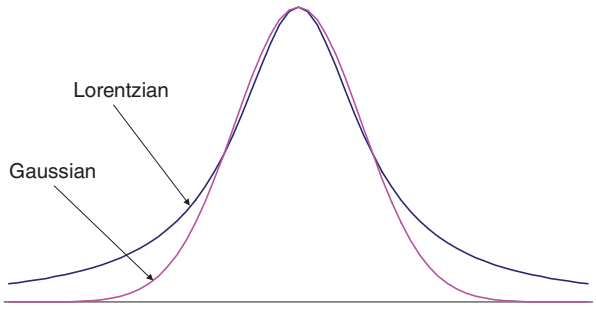


Figure 3.3 Gaussian and Lorentzian peak shapes of equal half heights.

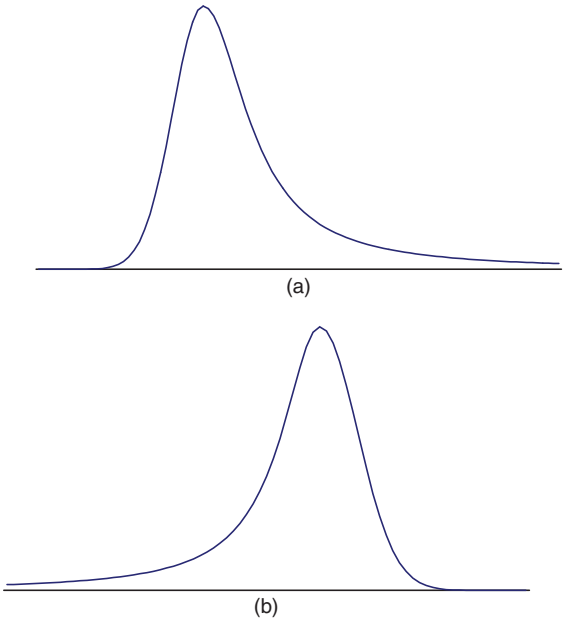


Figure 3.4 Asymmetric peak shapes often described by a Gaussian/Lorentzian model. (a) Tailing: left is Gaussian and right is Lorentzian (b) Fronting: left is Lorentzian and right is Gaussian.

represents a tailing peak shape, in which the left-hand side can be modelled by a Gaussian and the right-hand side by a Lorentzian. A fronting peak is illustrated in Figure 3.4(b): such peaks are much rarer.

1. The left-hand side of asymmetric peak is well modelled by a Lorentzian but the right-hand side by a Gaussian. The peak is
   (a) Tailing
   (b) Fronting

3.2.1.4 Use of Peak Shape Information

Peak shape information can be employed in two principal ways.

- *Curve fitting* is quite common. There are a variety of computational algorithms, most involving some type of least squares minimisation. If there are suspected (or known) to be three peaks in a cluster, of Gaussian shape, then nine parameters need to be found, namely the three peak positions, peak widths and peak heights. In any curve fitting, it is important to determine whether there is certain knowledge of the peak shapes, how many peaks are there and are of certain features, for example, the positions of each component, in advance. It is also important to appreciate that much chemical data are not of sufficient quality for very detailed models. In chromatography, an empirical approach is normally adequate: over-modelling can be dangerous. The result of the curve fitting can be a better description of the system; for example, by knowing peak areas, it may be possible to determine relative concentrations of components in a mixture. Note that if there is no prior knowledge of a system that could consist of a complex cluster of unresolved peaks, many curve fitting algorithms will fail; hence, it is usual to constrain the methods, for example, using known information to introduction restrictions to the model.
- Simulations also have an important role in chemometrics. Such simulations are a way of trying to understand a system. If the result of a chemometric method (such as multivariate curve resolution – see Chapter 7) results in reconstructions of peaks that are close to the real data, then the underlying peak shapes provide a good description. Simulations are also used to explore how well different techniques work and under what circumstances they break down.

A typical chromatogram or spectrum consists of several peaks, at different positions, of different intensities and sometimes of different shapes. Figure 3.5 represents a cluster of three peaks, together with their total intensity. Although the right-hand side peak pair is quite easy to resolve visually, this is not true for the left-hand side peak pair, and it would be especially hard to identify the position and intensity of the first peak of the cluster without using some form of data analysis.

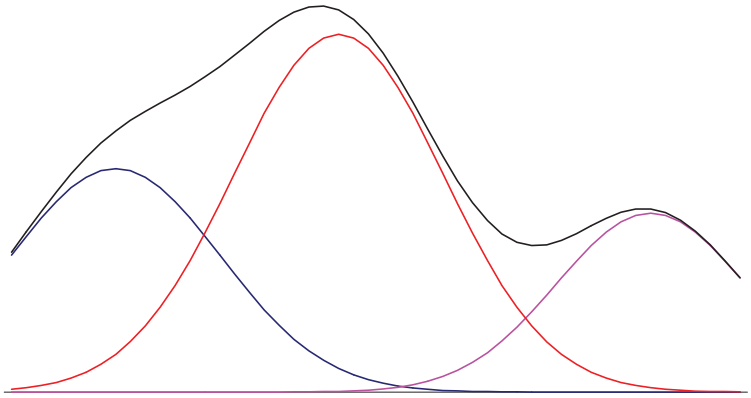


Figure 3.5 Three peaks forming a cluster.

1. Curve fitting is most appropriate when there is little or no information known about a cluster of partially overlapping peaks.
   (a) True
   (b) False

3.2.2 Digitisation

Almost all modern laboratory-based data are now obtained via computers and are acquired in a digitised rather than an analogue form. It is always important to understand how digital resolution influences the ability to resolve out peaks.

Many techniques for recording information result in only a small number of data points per peak. A typical NMR peak may be only a couple of hertz at half width, especially using well-resolved instrumentation. Yet, a spectrum recorded at 500 MHz, where 16 K (=16 384) data points are used to represent 10 ppm (or 5000 Hz), involves each data point representing 3.28 = 16 384/5000 Hz. A 2 Hz peak width is represented by only 6.56 data points. In coupled chromatography, a typical sampling rate may be 2 s; yet, peak half widths may be 20 s or less and interesting compounds separated by 30 s. Poor digital resolution can influence the ability to obtain information. It is useful to be able to determine how serious these errors are.

Consider a Gaussian peak, with a true width at half height of 16.65 units and a height of 1 unit. The theoretical area can be calculated using the equations of Section 3.2.1.1.

- The width at half height is given by $2s\sqrt{\ln 2}$, so that $s = 16.65/(2\sqrt{\ln 2}) = 10$.
- The area is given by $\sqrt{\pi}A\ s$, but $A = 1$, so that the area is $\sqrt{\pi}10 = 17.725\,\text{units}^2$.

Typical units might be AU * s if the sampling time is in s and the intensity in absorption units.

Consider the effect of digitising this peak at different rates, as indicated in Table 3.1 and illustrated in Figure 3.6. An easy way of determining integrated intensities is simply to sum the product of the intensity at each data point (x_i) by the sampling interval (δ) over a sufficiently wide range, that is, to calculate $\delta \sum x_i$. The estimates are given in Table 3.1, and it can be seen that for the worst digitised peak (at 20 units, or once per half height), the estimated integral is 20.157, an error of 13.7%.

A feature of Table 3.1 is that acquisition of data starts at exactly two data points in each case. In practice, the precise start of acquisition cannot easily be controlled and is often irreproducible, and it is easy to show that when poorly digitised, estimated integrals and apparent peak shapes will depend on this offset. In practice, the instrumental operator will notice a bigger variation in estimated integrals if digital resolution is low. Although peak widths must approach digital resolution for significant errors in integration, in some techniques such as gas chromatography mass spectrometry (GC-MS) or nuclear magnetic resonance (NMR), this condition is often obtained. In many situations, instrumental software is used to smooth or interpolate the data, and many users are unaware that this step has automatically taken place. These simple algorithms can result in considerable further distortions in quantitative parameters.

A second factor that can influence quantitation is digital resolution in the intensity and direction (or vertical scale in the graph). This is due to the analogue to digital converter (ADC) and sometimes can be experimentally corrected by

3.2.3 Noise

Imposed on signals is noise. In basic statistics, the nature and origin of noise are often unknown but assumed to obey a normal distribution. Indeed, many statistical tests such as the t-test and F-test (see Sections A.3.4 and A.3.5) assume this and are only approximations in the absence of experimental study of such noise distributions. However, a theorem called the *central limit theorem* suggests that most symmetrical distributions do tend towards a normal distribution in the centre, which is, therefore, a good approximation in the absence of other information.

In laboratory-based chemistry, there are two fundamental sources of noise in instrumental measurements.

- The first involves sample preparation, for example, dilution, weighing and extraction efficiency. We will not discuss these types of errors in this chapter, but can be minimised by good analytical procedures.
- The second is inherent to the measurement technique. No instrument is perfect; thus, the signal is imposed upon noise. The observed signal is given by

$$x = \tilde{x} + e$$

where $\tilde{x}$ is the 'perfect' or true signal and e represents noise. The aim of most signal-processing techniques is to obtain estimates of the true underlying signal in the absence of noise, that is, to separate the signal from the noise. The 'tilde' on top of the 'x' is to be distinguished from the 'hat' that refers to the estimated signal, often obtained from regression techniques including methods described in this chapter. Note that in this chapter, x will be used to denote the analytical signal or instrumental response, not y as in Chapter 2. This is to introduce a notation that is consistent with most of the open literature. Different investigators working in different areas of science often independently developed incompatible notation, and in an overview such as this text, it is preferable to stick reasonably closely to the generally accepted conventions to avoid confusion.

There are two main types of measurement noise.

3.2.3.1 Stationary Noise

The noise at each successive point (normally in time) does not depend on the noise at the previous point. In turn, there are two major types of stationary noise.

- *Homoscedastic noise*. This is the simplest to envisage. The features of the noise, normally the mean and standard deviation, remain constant over the entire data series. The most common type of noise is given by a normal distribution, usually with mean 0 unless there is a baseline problem, and standard deviation dependent on the instrument used. In most real-world situations, there are several sources of instrumental noise, but a combination of different symmetric noise distributions often tends towards a normal distribution.
- *Heteroscedastic noise*. This type of noise is dependent on signal intensity and is often proportional. The noise may still be represented by a normal distribution, but the standard deviation of that distribution is proportional to intensity of the underlying signal. A form of heteroscedastic noise often appears to arise if the data are transformed before processing, a common method being a logarithmic transform used in many types of spectroscopy such as UV/vis or IR spectroscopy, from transmittance to absorbance. The true noise distribution is imposed upon the raw data, but the transformed information distorts this.

Figure 3.7 illustrates the effect of both types of noise on a typical signal. It is important to recognise that several detailed models of noise are possible; however, in practice, it is not easy or interesting to perform sufficient experiments to determine such distributions. Indeed, it may be necessary to acquire several hundred or thousand spectra to obtain an adequate model, which represents overkill in most real-world situations. It is not possible to rely too heavily on published studies of instrumental noise distributions because each instrument is different and the experimental distribution is a balance between several sources, which differ in relative importance in each machine. In fact, as manufacture of instruments improves, certain types of noise are reduced in size and new effects come into play; hence, a thorough study of noise distributions performed a few decades ago is unlikely to be correct in detail on a more modern instrument. Many years ago, people studied instrumental noise distributions in more detail than is now necessary because the signal-to-noise ratios were lower, instruments changed less rapidly and they were often interested in improving the manufacture and performance of instruments.

In the absence of certain experimental knowledge, it is best to stick to a fairly straightforward distribution such as a normal distribution.

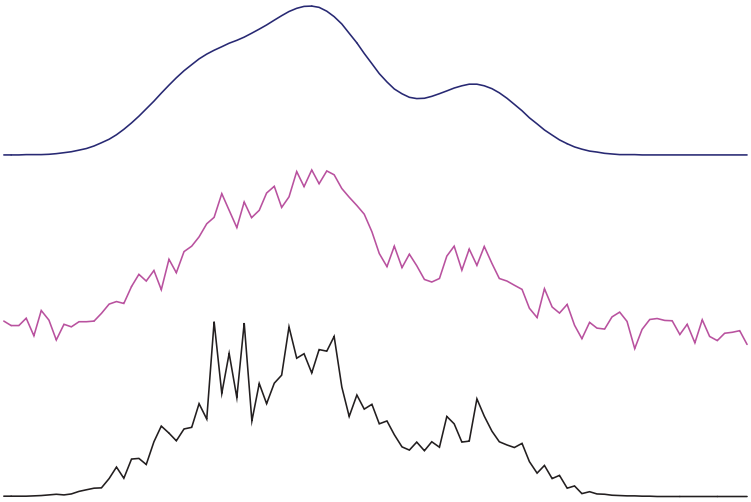


Figure 3.7 Examples of noise. From top to bottom: underlying signal, homoscedastic and heteroscedastic.

1. Heteroscedastic noise can be modelled by a normal distribution.
   (a) Never
   (b) Sometimes
   (c) Always

3.2.3.2 Correlated Noise

Sometimes, as a series is sampled, the level of noise in each sample depends on that of the previous sample. This is quite common in process control. For example, there may be problems in one aspect of the manufacturing procedure, an example being the proportion of an ingredient. If the proportion is in error by 0.5% at 2 in the afternoon, does this provide an indication of the error at 2.30?

Many such sources cannot be understood in great detail, but a generalised approach is that of *autoregressive moving average* (ARMA) noise.

- The *moving average* (MA) component relates the noise at time i to the values of the noise at previous times. A model of order p is given by $e_i = \sum_{t=0}^{t=p} c_{i-t} e_{i-t}$, where e_{i-t} is the noise at time $i-t$ and c_{i-t} a corresponding coefficient. A simple approach for simulating this type of noise is to put $p=1$ and set the coefficient to 1. Under such circumstances, $e_i = g_i + e_{i-1}$ where g_i is generated using a normal distribution. Table 3.2 illustrates a stationary noise distribution and a MA distribution generated by simply adding successive values of the noise, so that, for example, the noise at time = 4 is given by $-0.00927 = 0.05075 - 0.06001$.
- The *autoregressive* component relates the noise to the observed response at one or more previous times. A model of order p is given by $x_i = \sum_{t=0}^{t=p} c_{i-t} x_{i-t} + e_i$. Note that in a full ARMA model, e_i itself is dependent on past values of noise.

There is a huge literature on ARMA processes, which are particularly important in the analysis of long-term trends such as in economics: it is quite likely that an underlying factor causing errors in estimates changes with time rather than fluctuating completely randomly. There have been developed a battery of specialised techniques to cope with such situations. The chemist must be aware of these noise models, especially when studying natural phenomena such as in environmental chemistry but also to a lesser extent in instrumental analysis. However, there is rarely sufficient experimental evidence to establish highly sophisticated noise models. It is although well advised, when studying a process, to determine whether a stationary noise distribution is adequate, especially if the results of simulations are to be relied upon; thus, an appreciation of basic methods for modelling noise is important. Very elaborate models are unlikely to be easy to verify experimentally. In areas such as geology or economics, the nature of the underlying noise can be important for modelling, but it is not so essential in chemometrics, although it is useful to be aware of the existence of such models if studying natural phenomenon.

Table 3.2 Stationary and moving average noise.

Time	Stationary	Moving average
1	−0.12775	
2	0.14249	0.01474
3	−0.06001	−0.04527
4	0.05075	0.00548
5	0.06168	0.06716
6	−0.14433	−0.07717
7	−0.10591	−0.18308
8	0.06473	−0.11835
9	0.05499	−0.06336
10	−0.00058	−0.06394
11	0.04383	−0.02011
12	−0.08401	−0.10412
13	0.21477	0.11065
14	−0.01069	0.09996
15	−0.08397	0.01599
16	−0.14516	−0.12917
17	0.11493	−0.01424
18	0.00830	−0.00595
19	0.13089	0.12495
20	0.03747	0.16241

1. Heteroscedastic noise can be calculated by considering only the current measurement, whereas calculation of ARMA noise requires knowledge of past measurements.
   (a) True
   (b) False

3.2.3.3 Signal-to-Noise Ratio

The signal-to-noise ratio is a useful parameter to measure. The higher this number, the more intense the signal is relative to the background. This measurement is essentially empirical, and the most common definition involves dividing the height of a relevant signal (normally the most intense if there are several in a data set) by the root mean square of the noise, measured in a region of the data where there is known to be no signal. Most common measures of signal-to-noise ratio assume that the noise is homoscedastic. There usually needs to be an area of the signal where there are no peaks present and the baseline has to be subtracted.

If it is assumed that noise is normally distributed, it is possible to determine confidence that a 'blip' in a spectrum or chromatogram represents a peak or is just due to noise, according to the number of standard deviations it is above the baseline. If, for example, a 'blip' is 1.5 standard deviations above a baseline, 7.7% of measurements (see Table A.1) will exceed this level; hence, there is a 7.7% chance that the data point is due to random noise. Usually, either a 5% (1.65 standard deviation), a 1% (2.33 standard deviation) or a 0.1% (3 standard deviations) cut-off is used and the latter is sometimes called the *limit of detection*. Measurements more intense than these limits are considered to be real peaks rather than just artefacts of the noise.

1. The root mean square noise level is 2 and a data point has an intensity of 7 above the baseline. Noise is assumed to be homoscedastic and normally distributed.
   (a) There is more than 99% confidence that this represents a true peak rather than noise.
   (b) There is more than 95% but less than 99% confidence that this represents a true peak rather than noise.
   (c) There is insufficient confidence that this represents a true peak.

3.2.4 Cyclicity

Not all chemometric data arise from spectroscopy or chromatography; some are from studying processes evolving over time, ranging from a few hours (e.g. a manufacturing process) to thousands of years (e.g. a geological process). Many techniques for studying such processes are common to those developed in analytical instrumentation.

In some cases, cyclic events occur, dependent, for example, on time of day, season of the year or cyclical temperature fluctuations (such as diurnal fluctuations). These can be modelled using sine functions and are the basis of time series analysis (Section 3.4). In addition, cyclicity is also observed in Fourier spectroscopy, and Fourier transform techniques (Section 3.5) may occasionally be combined with methods for time series analysis.

1. Cyclical processes are often found in environmental chemistry.
   (a) True
   (b) False

3.3 Linear Filters

3.3.1 Smoothing Functions

A key need is to obtain a signal as informative as possible after removing the noise from a data set. When data are sequentially obtained, such as in time or frequency, the underlying signals often arise from a sum of smooth, monotonic functions, such as those described in Section 3.2.1, whereas the underlying noise is often an uncorrelated function. An important method for revealing the signals involves smoothing the data; the principle is that the noise will be smoothed away using quite mild methods, whilst the signal, being broader, will remain. This approach depends on the peaks having a half width of several data points: if digital resolution is very poor, signals may appear as spikes and be confused with noise.

It is important to determine the optimum filter for any particular application. Too much smoothing and the signal itself is reduced in intensity and resolution. Too little smoothing and noise remains. The optimum smoothing function depends on peak widths (in data points) as well as noise characteristics.

3.3.1.1 Moving Averages

Conceptually, the simplest methods are linear filters where the resultant smoothed data are given as a linear function of the original data. Normally, this involves using the surrounding data points, for example, using a function of the three points in Figure 3.8, to recalculate a value for the central point i. Algebraically, such functions are expressed by

$$x_{i,new} = \sum_{j=-p}^{p} c_j x_{i+j}$$

One of the simplest is a three-point MA. Each point is replaced by the average of itself and the points before and after, so in the above equation,

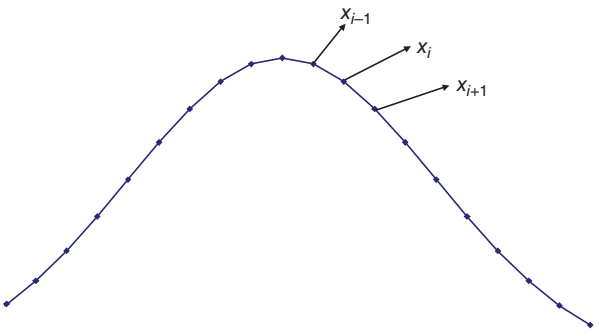


Figure 3.8 Selection of points to be used in a three-point moving average filter.

- $p = 1$
- $c_j = 1/3$ for all three points.

The filter can be extended to a five-point MA ($p = 2$, $c = 1/5$), seven-point MA and so on.

- The more the points in the filter, the greater the reduction in noise, but the higher the chance of blurring the signal.
- The number of points p in the filter is often called a *window* or a *filter width*.

The filter is moved along the time series or spectrum, each data point being replaced successively by the corresponding filtered data point. The optimal filter depends on the noise distribution and signal width. It is best to experiment with a number of different filter widths to find the optimum. The $(p-1)/2$ points at the beginning and end of a series are usually removed; for example, if using a five-point window, points 1–2 are removed, and the first new filtered point is number 3.

1. A time series consists of 80 data points. It is filtered using a seven-point moving average. The resultant time series consists of
   (a) 77 points
   (b) 74 points
   (c) 68 points

3.3.1.2 Savitzky–Golay Filters, Hanning and Hamming Windows

MA filters have the disadvantage that they use a linear approximation for the data. However, peaks are often best approximated by polynomial curves. This is particularly true at the centre of a peak, where a linear model will always underestimate the intensity. Quadratic, cubic or even quartic models provide better approximations. The principle of MAs can be extended to polynomials. A seven-point cubic filter, for example, is used to fit a model

$$x_i = b_0 + b_1 i + b_2 i^2 + b_3 i^3$$

using a seven-point window, replacing the centre point by its best-fit estimate. The window is moved along the data, point by point, the calculation being repeated each time.

However, regression is computationally intense, and it would be computationally slow to perform this calculation in full to simply improve the appearance of a spectrum or chromatogram, which may consist of thousands of data points. The user wants to be able to select a menu item or icon on a screen and almost instantaneously visualise an improved picture. Savitzky and Golay in 1964 presented an alternative and simplified method of determining the new value of x_i simply by re-expressing the calculation as a sum of coefficients. These Savitzky–Golay filters are normally represented in a tabular form (see Table 3.3). To determine a coefficient c_j,

Table 3.3 Savitzky–Golay coefficients c_{i+j} for smoothing.

Window size	5	7	9	7	9
j	Quadratic/cubic			Quartic/quintic	
−4			−21		15
−3		−2	14	5	−55
−2	−3	3	39	−30	30
−1	12	6	54	75	135
0	17	7	59	131	179
1	12	6	54	75	135
2	−3	3	39	−30	30
3		−2	14	5	−55
4			−21		15
Normalisation constant	35	21	231	231	429

- decide on the order of the model (quadratic and cubic models give identical results as do quartic and quintic models),
- decide on the window size and
- determine c_j by selecting the appropriate number from Table 3.3 and dividing by the corresponding normalisation constant.

Several other MA methods have been proposed in the literature, two of the best known being the Hanning window (named after Julius Von Hann) (for which three points has weights 0.25, 0.5 and 0.25) and the Hamming window (named after R.W. Hamming) (for which five points has weights 0.0357, 0.2411, 0.4464, 0.2411 and 0.0357) – not to be confused but very similar in effects. These windows can be calculated for any size, but we recommend these two basic filter sizes.

Notice that although quadratic, cubic or higher approximations of the data are employed, the filters are still called linear because each filtered point is a linear combination of the original data.

1. A linear filter is defined as
   (a) A filter that uses linear combinations of original data to obtain a local model of a sequential series even though the model may be polynomial.
   (b) A filter that always results in a linear model of a sequential series.

3.3.1.3 Calculation of Linear Filters

The calculation of MA and Savitzky–Golay filters is illustrated in Table 3.4.

- The first point of the three-point MA (see column 2) is simply given by

$$-0.049 = (0.079 - 0.060 - 0.166)/3$$

- The first point of the seven-point Savitzky–Golay quadratic/cubic filtered data can be calculated as follows.
  From Table 3.3, obtain the seven coefficients, namely $c_{-3} = c_3 = -2/21 = -0.095$, $c_{-2} = c_2 = 3/21 = 0.143$ and $c_{-1} = c_1 = 6/21 = 0.286$, $c_0 = 7/21 = 0.333$.
  Multiply these coefficients by the raw data and sum to get the smoothed value of the data

$$\begin{aligned} x_{i,new} &= -0.095 \times 0.079 + 0.143 \times -0.060 + 0.286 \times -0.166 + 0.333 \times -0.113 \\ &\quad + 0.286 \times 0.111 + 0.143 \times 0.145 - 0.095 \times 0.212 \\ &= -0.069 \end{aligned}$$

Figure 3.9(a) is a representation of the raw data. The result of using MA filters is shown in Figure 3.9(b). A three-point MA preserves the resolution (just), but a five-point MA loses this and the cluster appears to be composed of only one peak. In contrast, the five- and seven-point quadratic/cubic Savitzky–Golay filters (Figure 3.9(c)) preserve resolution whilst reducing noise and only starts to lose resolution when using a nine-point function.

1. Seven successive data points are as follows:

−0.008	0.299	0.410	0.361	0.175	0.101	0.184

   Using a seven-point cubic Savitzky–Golay filter, the central point after smoothing is
   (a) 0.120
   (b) 0.307
   (c) 0.328
   (d) 6.891

3.3.1.4 Running Median Smoothing

Most conventional filters involve computing local multi-linear models; however, in certain areas such as process analysis, there can be spikes (or outliers) in the data, which are unlikely to be part of a continuous process. An alternative method involves using RMS functions, which calculate the median rather than the mean over a window. An example of a process is given in Table 3.5. A five-point MA and five-point RMS smoothing function is compared. A check on

sudden glitch in the system unlikely to be a real effect. Often, a combination of filters is recommended, for example, a five-point median smoothing followed by a three-point Hanning window. These methods are very easy to implement computationally, and it is possible to view the results of different filters simultaneously.

1. Running median smoothing can result in flat peak shapes.
   (a) True
   (b) False

3.3.2 Derivatives

The methods described in Section 3.3.1 are primarily concerned with removing noise. Most such methods either leave peak widths unchanged or increased, equivalent to blurring. In signal analysis, an important separate need is to increase resolution. In Section 3.5.2, we will discuss the use of filters combined with Fourier transformation. In Chapter 7, we

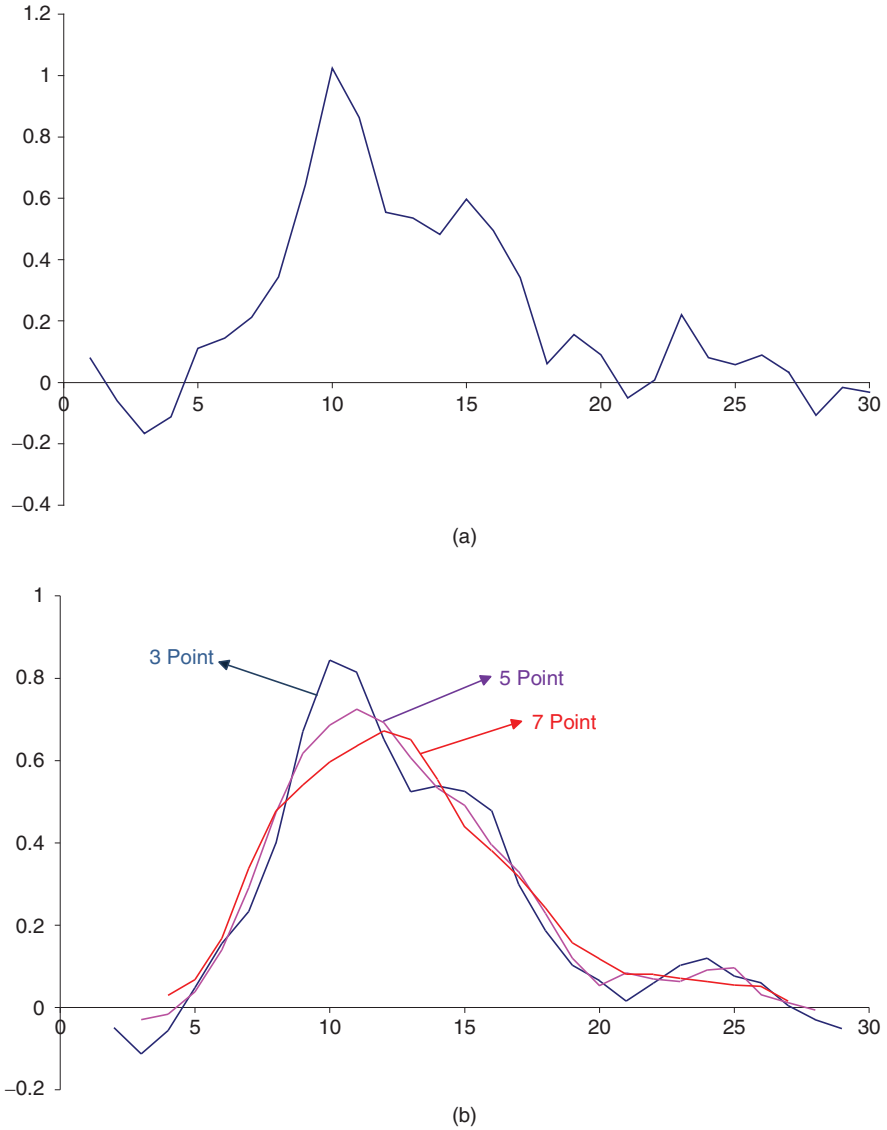


Figure 3.9 Filtering of data. (a) Raw data, (b) moving average filters, (c) quadratic/cubic Savitzky–Golay filters.

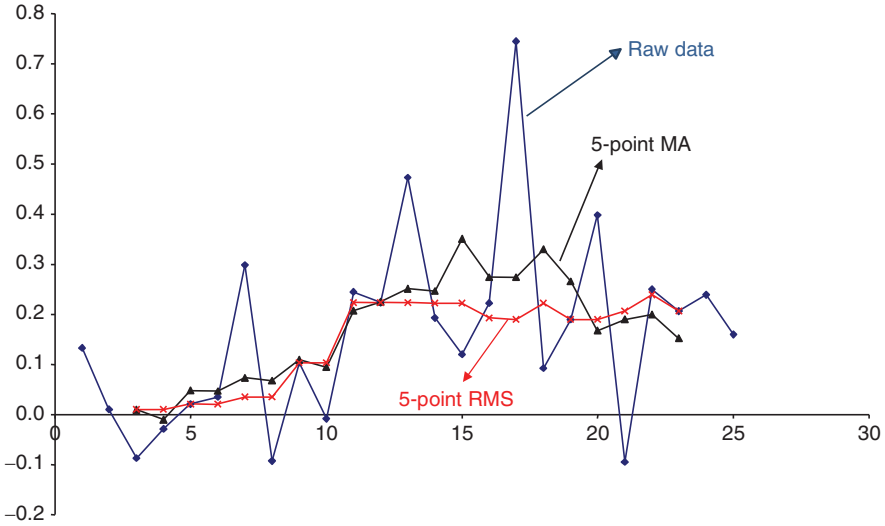


Figure 3.10 Comparison of moving average and running median smoothing.

will discuss how to improve resolution when there is an extra dimension to the data (e.g. multivariate curve resolution). However, a simple and frequently used approach is to calculate derivatives. The principle is that inflection points in partially resolved peaks become turning points in the derivatives. The first and second derivatives of a pure Gaussian are presented in Figure 3.11.

- The first derivative equals zero at the centre of the peak and is a good way of accurately pinpointing the position of a broad peak. It exhibits two turning points.
- The second derivative is a minimum at the centre of the peak, crosses zero at the positions of the turning points for the first derivative and exhibits two further turning points farther apart than in the first derivative.
- The apparent peak width is reduced using derivatives.

The properties are most useful when there are several closely overlapping peaks, and higher order derivatives are often employed, for example in electron spin resonance and electronic absorption spectroscopy to improve resolution. Figure 3.12 illustrates the first and second derivatives of two closely overlapping peaks. The second derivative clearly indicates two peaks and allows one to pinpoint their positions. The first derivative would suggest that the peak is not pure but, in this case, probably does not provide definitive evidence. It is, of course, possible to continue and calculate the third derivative, fourth derivative and so on, but the patterns can become quite complicated.

There are, however, two disadvantages of using derivatives. First, they can be computationally intense, as a fresh calculation is required for each data point in a spectrum or chromatogram. Second, and most importantly, they amplify noise substantially, and, therefore, require low signal-to-noise ratios. These limitations can be overcome by using Savitzky–Golay coefficients similar to those described in Section 3.3.1.3, which involve rapid calculation of smoothed higher derivatives. The coefficients for a number of window sizes and approximations are presented in Table 3.6. This is a common method for the determination of derivatives and implemented in many software packages.

1. Savitzky–Golay derivative functions are computationally rapid methods for calculating derivatives but they amplify noise.

   (a) True
   (b) False

3.3.3 Convolution

Common principles occur in different areas of science, often under different names and are introduced in conceptually different guises. In many cases, the driving force is the expectations of readers, who may be potential users of techniques, customers on courses or programmers, often with a variety of backgrounds such as engineering, analytical

1. A Savitzky–Golay filter can be considered a convolution function.
   (a) True
   (b) False

3.4 Correlograms and Time Series Analysis

Time series analysis has a long statistical vintage, with major early applications in economics and engineering. The aim is to study cyclical trends in processes. In the methods of Section 3.3, we were mainly concerned with peaks arising from chromatography or spectroscopy or else processes that occur in manufacturing. There were no underlying cyclical features. However, in certain circumstances, features can re-occur at regular intervals. These could arise from a geological process, a manufacturing plant or environmental monitoring, the cyclic changes being due to season of the year, time of day or even hourly events.

The aim of time series analysis is to mainly reveal the cyclical trends in a data set. These will be buried within non-cyclical phenomena and also various sources of noise. In spectroscopy, where the noise distributions are well understood and primarily stationary, Fourier transforms are the method of choice. However, when studying natural processes, there are likely to be a much larger number of factors influencing the response, including often correlated (or ARMA) noise, as discussed in Section 3.2.3.2. Under such circumstances, time series analysis is preferable and can reveal quite weak cyclicity. The disadvantage is that original intensities are lost, the resultant information being primarily about how strong the evidence is that a particular process exhibits cyclicity. Most methods for time series analysis involve the calculation of a correlogram at some stage.

3.4.1 Auto-correlograms

Consider the information depicted in Figure 3.14, which represents a process changing with time. It appears that there is some cyclicity, but this is buried within the noise. The numerical data are presented in Table 3.7.

An auto-correlogram involves calculating the correlation coefficient between a time series and itself, shifted by a given number of data points called a 'lag'. If there are I data points in the original time series, then a correlation coefficient for a lag of l points will involve $I - l$ data points. Hence, in the case of Table 3.7, there are 30 points in the original data set but only 25 points in the data set for which $l = 5$. Point number 1 in the shifted data set corresponds to point number 6 in the original data set. A common equation for the correlation coefficient for a time series lagged by l is given by

$$r_l = \frac{\sum_{i-1}^{I-l}(x_i - \bar{x})(x_{i+l} - \bar{x})}{\sum_{i=1}^{I}(x_i - \bar{x})^2}$$

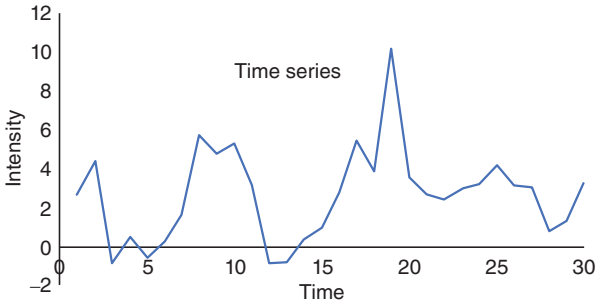


Figure 3.14 A time series.

in which it is assumed that the mean and variance (or standard deviation) vary according to the amount of lag. In practice, unless the lag is large relative to the time series, there is very little difference between these two equations, and below we will use the simpler equation.

In addition, there are a number of other computational alternatives to these two formulae.

There are a number of properties of the correlogram.

- For a lag of 0, the correlation coefficient is 1.
- It is possible to have negative lags as well as positive lags, but for an auto-correlogram, $r_l = r_{-l}$ and sometimes only one half of the correlogram is displayed.
- The closer the correlation coefficient is to 1, the more similar the two series. If a high correlation is observed for a large lag, this indicates cyclicity.
- As the lag increases, the number of data points used to calculate the correlation coefficient decreases; hence, r_l becomes less informative and more dependent on noise. Large values of l are not advisable, a good compromise is to calculate the correlogram for values of l between half or two-thirds of I.

The resultant auto-correlogram for the data in Table 3.7 is presented in Figure 3.15. The cyclic pattern is now much clearer than in the original data. Note that the graph is symmetric about the origin as expected, and the maximum lag used in this example equals 20 points, being 30 points in the original data.

An auto-correlogram emphasises only cyclical features. Sometimes, there are non-cyclical trends superimposed over the time series. Such situations regularly occur in economics. Consider trying to determine the factors relating to expenditure in a seaside resort. A cyclical factor will undoubtedly be seasonal, being more business in the summer. However, other factors such as interest rates, exchange rates and long-term economic activity will also come into play and the information will be mixed up in the resultant statistics. Expenditure could also be divided into food, accommodation, clothes and so on. Each will be influenced to different extent by seasonality. Correlograms specifically emphasise the cyclical causes of expenditure. In chemistry, they are most valuable when time-dependent noise interferes with stationary noise, for example in the environmental chemistry of a river where there may be specific types of pollutants or changes in chemicals that occur spasmodically but once discharged take time to dissipate as well, for example, in diurnal cyclical factors.

The correlogram can be further processed by Fourier transformation, smoothing functions or a combination of both, whose techniques are discussed in Sections 3.3 and 3.5. Sometimes, the results can be represented in the form of probabilities, for example, the chance that there is really a genuine underlying cyclical trend of a given frequency. Such calculations, although, make certain definitive assumptions about the underlying noise distributions and experimental error and cannot always be generalised and would be rare in chemistry, although there is a large field of time series analysis that is well developed in geology, economics and so on.

1. Auto-correlograms are always symmetrical around zero lag.
   (a) True
   (b) False
2. A time series consists of 40 data points. An auto-correlogram is calculated for lag up to 25, including negative lags. The total number of points in the correlogram is
   (a) 40
   (b) 50
   (c) 51
   (d) 81

3.4.2 Cross-correlograms

It is possible to extend these principles to the comparison of two independent time series. Consider measuring the levels of Ag and Ni in a river with time. Although each may show a cyclical trend, are there trends common to both metals? The cross-correlation function between two series represented by x and y can be calculated for a lag of l,

$$r_l = \frac{c_{xy,l}}{s_x s_y}$$

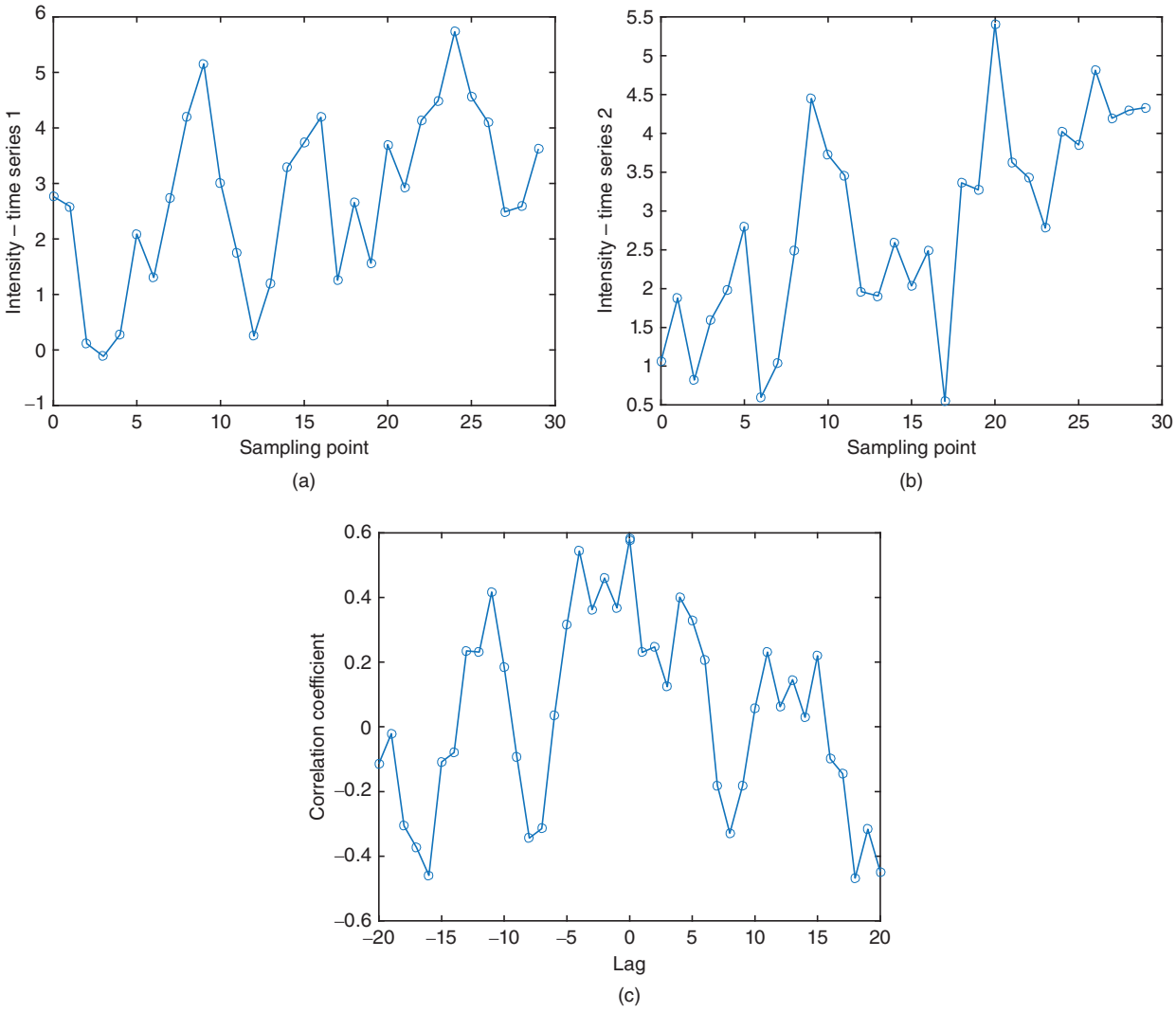


Figure 3.16 Two time series (a,b) and their corresponding cross-correlogram (c).

1. Cross-correlograms are always symmetrical around zero lag.
   (a) True
   (b) False

3.4.3 Multivariate Correlograms

In the real-world, there may be a large number of variables that change with time, for example, the composition of a manufactured product. In a chemical plant, the resultant material could depend on a huge number of factors such as the quality of the raw material, the performance of the apparatus, even the time of day which could relate to who is on shift, or small changes in power supplies. Instead of monitoring each factor individually, it is common to obtain an overall statistical indicator, typically the scores of a principal component (see Chapter 4), but sometimes other statistical indicators such as the Mahalanobis distance from the centroid of a batch of typical in control or acceptable

samples are also obtained (see Chapter 5). The correlogram is computed from this mathematical summary of the raw data rather than the concentration of an individual constituent.

1. Principal component scores can show cyclicity.
   (a) True
   (b) False

3.5 Fourier Transform Techniques

The mathematics of Fourier transformation has been well established for two centuries, but early computational algorithms were first described in the 1960s, a prime method being the Cooley–Tukey algorithm. Originally employed in physics and engineering, Fourier transform (FT) techniques are now essential tools of the chemists. Modern NMR, IR and X-ray spectroscopy, among others, depend on Fourier transform methods. FTs have been extended to two-dimensional time series, plus a wide variety of modifications, for example, phasing, resolution enhancement and applications to image analysis have been developed over the past decades. For certain types of instrumentation, most notably NMR, there is a vast literature on enhancements to FT methods, but below we primarily focus on the main approaches, allowing the reader to delve into the specialist literature where appropriate.

3.5.1 Fourier Transforms

3.5.1.1 General Principles

The original literature on Fourier series and transform techniques involved primarily applications to continuous data sets. However, in chemical instrumentation, data are not sampled continuously but at regular intervals of time, so all data are digitised. The discrete Fourier transform (DFT) is used to process such data and will be described below. It is important to recognise that DFTs have specific properties that distinguish them from continuous FTs.

DFTs involve transformation between two types of data. In FTNMR, the raw data are acquired at regular intervals of time, often called the *time domain* or more specifically described by a Free Induction Decay (FID). FTNMR has been developed over the years because it is much quicker to obtain data than using conventional (continuous wave) methods. An entire spectrum can be sampled in a few seconds, rather than minutes, speeding up the procedure of data acquisition by one or two orders of magnitude. This has meant that it is possible to record spectra of small quantities of compounds or of natural abundance of isotopes such as ^{13}C, now routine in modern laboratories.

The trouble with this is that the time domain is not easy to interpret, and here arises the need for DFTs. Each peak in a spectrum can be described by three parameters, namely a height, width and position, as discussed in Section 3.2.1. In addition, each peak has a shape; in NMR, this is Lorentzian. A spectrum consists of a sum of peaks and is often referred to as the *frequency domain*. However, raw data, for example, in NMR, are recorded in the *time domain* and each frequency domain peak corresponds to a time series characterised by

- an initial intensity,
- an oscillation rate and
- a decay rate.

The time domain consists of a sum of time series, each corresponding to a peak in the spectrum. Noise is superimposed on this time series. Fourier transforms convert the time series into a recognisable spectrum as indicated in Figure 3.17. Each parameter in the time domain corresponds to a parameter in the frequency domain as indicated in Table 3.9.

- The faster the rate of oscillation in the time series, the farther away the peak is from the origin in the spectrum.
- The faster the rate of decay in the time series, the broader the peak in the spectrum.
- The higher the initial intensity in the time series, the greater the area of the transformed peak.

The peak shape in the frequency domain relates to the decay curve (or mechanism) in the time domain. The time domain equivalent to a Lorentzian peak is

$$f(t) = A\cos(\omega t)e^{-t/s}$$

spectroscopy is that all the peaks can be observed simultaneously, thus allowing rapid acquisition of data, but a mathematical transform is required to make the data comprehensible.

1. The broader a peak is in a spectrum, the faster its corresponding time domain profile will decay.
   (a) True
   (b) False

3.5.1.2 Fourier Transform Methods

The process of Fourier transformation converts the raw data (e.g. a time series) into two frequency domain spectra, one of which is called a *real spectrum* and the other called *imaginary* (this terminology comes from complex numbers). The true spectrum is represented only by half the transformed data, as indicated in Figure 3.19. Hence, if there are 1024 data points in the original time series, 512 will correspond to the real transform and 512 to the imaginary transform.

The mathematics of Fourier transformation is not too difficult to understand, but it is important to realise that authors use slightly different terminology, and definitions, especially with regard to constants in the transform. When reading a paper or text, consider these factors very carefully and always check that the result is realistic. We will adopt a number of definitions as follows.

The *forward* transform converts a purely *real* series into both a real and an imaginary transform, whose spectrum may be defined by

$$\mathrm{F}(\omega) = \mathrm{RL}(\omega) - i\ \mathrm{IM}(\omega)$$

where F is the Fourier transform, ω is the frequency in the spectrum, i the square root of -1 and RL and IM the two halves of the transform.

The *real* part is obtained by performing a *cosine* transform on the original data, given by (in its simplest form)

$$\mathrm{RL}(n) = \sum_{m=0}^{M-1} f(m) \cos(nm/M)$$

and the *imaginary* part by performing a *sine* transform

$$\mathrm{IM}(n) = \sum_{m=0}^{M-1} f(m) \sin(nm/M)$$

These are sometimes expressed as $\mathrm{F}(n) = \sum_{n=0}^{M-1} \exp(-i\pi nm/M)$, where i is the square root of -1 and $\mathrm{F}(n) = \mathrm{RL}(n) - i\ \mathrm{IM}(n)$ (there is a relationship for complex exponentials called *Euler's formula*).

These terms need some definition.

- There are M data points in the original (time series) data,
- m refers to each point in the time series,
- n is a particular point in the transform, usually, in our case, a spectrum and
- the angles are in cycles per second.

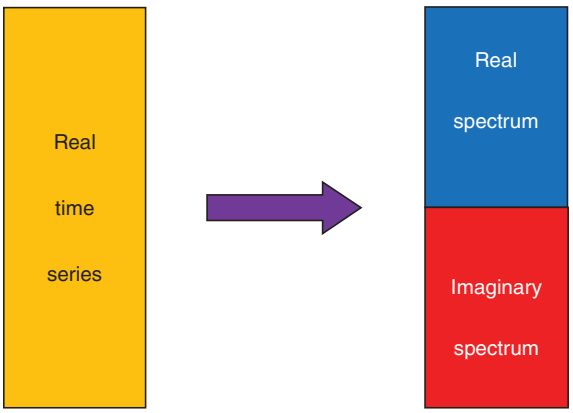


Figure 3.19 Transformation of a real time series to real and imaginary pairs.

If you use radians, you must multiply the angles by 2π and if degrees by 360°, but the equations above are presented in a simple way. There are quite a number of methods for determining the units of the transformed data, provided we are transforming a purely real time series to a real spectrum of half the size ($M/2$), then if the sampling interval in the time domain is δt s, the interval of each data point in the frequency domain is $\delta\omega = 1/(M\ \delta t)$ Hz (= cycles per second). To give an example, if we record 8000 data points in total in the time domain at intervals of 0.001 s (so the total acquisition time is 8 s), then the real spectrum will consist of 4000 data points at intervals of $1/(8000 \times 0.001) = 0.125$ Hz, thus will cover a total range of detectable frequencies over $4000 \times 0.125 = 500$ Hz. The rationale behind these numbers will be described in Section 3.5.1.4. Some books contain equations that appear more complicated to those presented here because they transform from time to frequency units rather than that from data points.

An *inverse* transform converts the real and imaginary pairs into a real series and is of the form

$$f(t) = \mathrm{rl}(t) + i\ \mathrm{im}(t)$$

Notice the '+' sign. Otherwise, the transform is similar to the forward transform, the real part involving the multiplication of a cosine wave with the spectrum. Sometimes, a factor of $1/N$, where there are N data points in the transformed data, is applied to the inverse transform, so that a combination of forward and inverse transforms gives the original time series.

FTs are best understood by a simple numerical example. For simplicity, we will give an example where there is a purely real spectrum and both real and imaginary time series – the opposite to normal but perfectly reasonable: in the case of Fourier self-convolution (Section 3.5.2.3), this indeed is the procedure. We will show only the real half of the transformed time series. Consider a spike as shown in Figure 3.20. The spectrum is of zero intensity except at one point, $m = 2$. We assume that there are M (=20) points numbered from 0 to 19 in the spectrum.

What happens to the first 10 points of the transform? The values are given by

$$\mathrm{RL}(n) = \sum_{m=0}^{19} f(m) \cos(nm/M)$$

As $f(m)$ equals 0 except where $m = 2$, when it equals 10, the equation simplifies furthermore so that

$$\mathrm{RL}(n) = 10\ \cos(2n/20)$$

The angular units of the cosine are cycles per unit time; hence, this angle must be multiplied by 2π to convert into radians. (While employing computer packages for trigonometry, always check whether units are in degrees, radians or cycles: this is simple to do; the cosine of 360° equals the cosine of 2π radians, which equals the cosine of 1 cycle and equals 1.) As shown in Figure 3.20, there is half a cycle every 10 data points, as $2 \times 10/20 = 1$, and the initial intensity equals 10 because this is the area of the spike (obtaining by summing the intensity in the spectrum over all data points). It should be evident that the farther the spike from the origin, the more the cycles in the transform. Similar calculations can be employed to demonstrate other properties of Fourier transforms as discussed above.

1. A time series consists of 5000 data points, acquired at regular intervals of 0.002 s. The range of detectable spectral frequencies is
   (a) 200 Hz
   (b) 250 Hz
   (c) 500 Hz
2. A spectrum consists of a spike at data point 5. If the spectrum is real and recorded over 50 data points, the resultant real transform consists of a cosine function with the following number of maxima.
   (a) 2
   (b) 3
   (c) 5
   (d) 6

3.5.1.3 Real and Imaginary Pairs

In the Fourier transform of a real time series, the peak shapes in the real and imaginary halves of the spectrum differ. Ideally, the real spectrum corresponds to an *absorption* line shape and the imaginary spectrum to a *dispersion* line

There are a variety of solutions to this problem, a common one being to correct this by adding together proportions of the real and imaginary data until an absorption peak shape is achieved using an angle ϕ, so that

$$\mathrm{ABS} = \cos(\phi)\,\mathrm{RL} + \sin(\phi)\,\mathrm{IM}$$

Ideally, this angle should equal the phase angle, which is usually experimentally unknown. Sometimes, phasing is fairly tedious experimentally and can change across a spectrum, and there are, in addition to manual methods, a number of algorithms for phasing. For complex problems such as two-dimensional Fourier transforms, phasing can be quite difficult.

An alternative is to take the absolute value, or magnitude, spectrum defined by

$$\mathrm{MAG} = \sqrt{\mathrm{RL}^2 + \mathrm{IM}^2}$$

The power spectrum is sometimes also used, which is the square of the magnitude spectrum. Although the magnitude spectrum is easy to calculate and always positive, it is important to realise that it is not quantitative: the peak area of a two-component mixture is not equal to the sum of peak areas of each individual component, the reason being that the sum of squares of two numbers is not equal to the square of their sum. As, sometimes, spectroscopic peak areas (or heights) are used for chemometric pattern recognition studies, this limitation is important to appreciate, although it is not important in, for example, imaging.

1. For a phase angle of −90°, the real spectrum corresponds to the absorption spectrum and the imaginary to the dispersion spectrum.
   (a) True
   (b) False

3.5.1.4 Sampling Rates and Nyquist Frequency

An important property of DFTs relates to the rate which data are sampled. Consider the time series in Figure 3.23, each square indicating a sampling point. If it is sampled at half the rate, it will appear that there is no oscillation, as every alternative data point will be eliminated. Therefore, there is no way of distinguishing such a series from a zero frequency series. The oscillation frequency in Figure 3.23 is called the *Nyquist frequency*. Anything that oscillates faster than this frequency will be indistinguishable from a sine wave at a lower frequency. The rate of sampling establishes the range of observable frequencies. The higher the sampling rate in time, the greater the range of observable frequencies. In order to increase the spectral width, a higher sampling rate is required; hence, more data points must be collected per unit time. The equation

$$M = 2\,S\,T$$

links the number of data points acquired (e.g. $M = 4000$) in the time domain, the range of observable frequencies (e.g. $S = 500\,\mathrm{Hz}$) and the acquisition time (e.g. $T = 4\,\mathrm{s}$). Higher frequencies are said to be 'folded over' or 'aliasing'

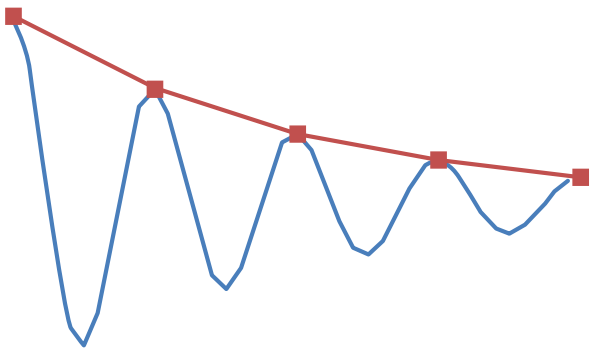


Figure 3.23 A sparsely sampled time series sampled at the Nyquist frequency. Blue: underlying time series, red observed time series if sparsely sampled.

and appear to be at lower frequencies, as they are undistinguishable. If $S = 500$ Hz, a peak oscillating at 600 Hz will appear indistinguishable from a peak at 400 Hz in the transform. Notice that this relationship determines how a sampling rate in the time domain results in a digital resolution in the frequency or spectral domain (see Section 3.5.1.2). In the time domain, if samples are taken every $\delta t = T/M$ s, in the frequency domain, we obtain a data point every $\delta\omega = 2S/M = 1/T = 1/(M\,\delta t)$ Hz in the spectral domain. Notice that in certain spectroscopies (such as quadrature detection FTNMR), it is possible to record both two time domain signals (treated mathematically as real and imaginary time series) and transform these into real and imaginary spectra. In such cases, only $M/2$ points are recorded in time, so the sampling frequency in the time domain is halved.

The Nyquist frequency is not only important in instrumental analysis. Consider sampling a geological core, where depth relates to time, to determine whether the change in concentrations of a compound, or isotopic ratios, exhibits cyclicity. A finite amount of core is needed to obtain adequate quality samples, which means there is a limitation in samples per length of core. This, in turn, limits the maximum frequency that can be observed. More intense sampling may require a more sensitive analytical technique; thus, for a given method, there is a limitation to the range of frequencies that can be observed.

Note that slightly different terminologies are used for different techniques, but the description above is based on NMR. The units for the range of spectral frequencies are expressed as hertz (or cycles per second) for the equations used above, but of course must be changed as appropriate.

1. A spectrum consists of 2048 real points in the frequency domain and a 1000 Hz range of observable frequencies. What was the acquisition time?
   (a) 2.048 s
   (b) 1.024 s

3.5.1.5 Fourier Algorithms

A final consideration relates to algorithms used for Fourier transforms. DFT methods became widespread in the 1960s partly because Cooley and Tukey developed a rapid computational method, the fast Fourier transform (FFT). This method required the number of sampling points to be a power of 2, for example, 1024, 2048 and so on, and many chemists still associate powers of 2 with Fourier transformation. However, there is no special restriction on the number of data points in a time series, the only consideration is relating to the speed of computation. The method for Fourier transformation introduced above is slow for large data sets, and early computers were much more limited in capabilities, but it is not always necessary to use rapid algorithms in modern day applications unless the amount of data is really large. There is a huge technical literature on Fourier transform algorithms, but it is important to recognise that an algorithm is simply a means to an end, and not an end in itself.

1. It is necessary for the number of data points in a time series to be a power of 2 when using the Cooley–Tukey algorithm for Fourier transformation.
   (a) True
   (b) False

3.5.2 Fourier Filters

In Section 3.3, we discussed a number of linear filter functions that can be used to enhance the quality of spectra and chromatograms. When performing Fourier transforms, it is possible to apply filters to the raw (time domain) data before Fourier transformation, and this is a common method in spectroscopy to enhance resolution or signal-to-noise ratio, as an alternative to applying filters directly to the spectral domain. There is a huge literature on this; thus, we only describe some of the most common methods below.

3.5.2.1 Exponential Filters

The width of a peak in a spectrum primarily depends on the decay rate in the time domain. The faster the decay, the broader the peak. Figure 3.24 illustrates a broad peak together with its corresponding time domain. If it is desired to increase resolution, a simple approach could be to change the shape of the time domain function so that the decay is

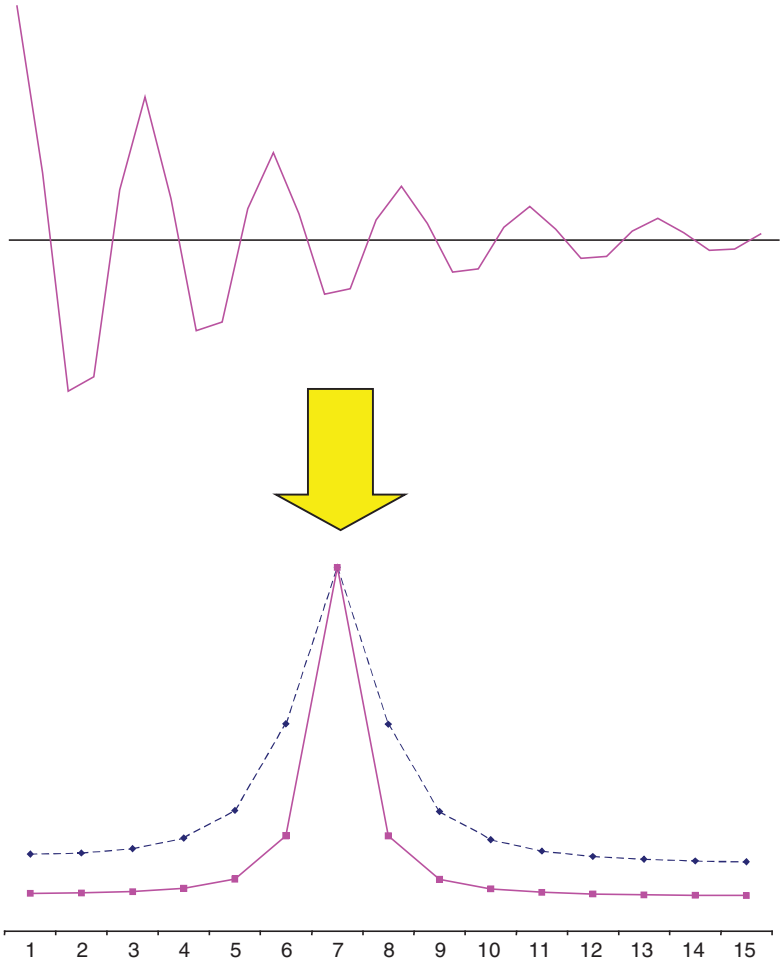


Figure 3.25 Result of multiplying the time series in Figure 3.24 by a positive exponential, the transform of the original time series being represented by a dotted blue line.

3.5.2.2 Influence of Noise

Theoretically, it is possible to conceive multiplying the original time series by increasingly positive exponentials until peaks are one data point wide. Clearly, there is a flaw in our argument; otherwise, it would be possible to obtain indefinitely narrow peaks and achieve any desired resolution.

The difficulty is that real spectra always contain noise. Figure 3.26 represents a noisy time series, together with the exponentially filtered data. The filtered time series amplifies noise substantially, which can interfere with signals. Although the peak width of the new transform has indeed decreased, the noise has increased. In addition to making peaks hard to identify, noise also reduces the ability to determine integrals and concentrations and sometimes to accurately pinpoint peak positions.

How can this be solved? Clearly, there are limits to the amount of peak sharpening that is practicable, but the filter function can be improved so that noise reduction and resolution enhancement are applied simultaneously. One common method is to multiply the time series by a double exponential filter of the form

$$g(t) = e^{+\kappa t - \nu t^2}$$

where the first (linear) term of the exponential increases with time and enhances resolution, and the second (quadratic) term decreases noise. Provided the values of κ and ν are chosen correctly, the result will be an increase in resolution without an increase in noise. The main aim is to emphasise the middle of the time series whilst reducing the end. These two terms could be optimised theoretically if peak widths and noise levels are known in advance, but, in most practical cases, they are chosen empirically. The effect on the noisy data in Figure 3.26 is illustrated in Figure 3.27, for a typical

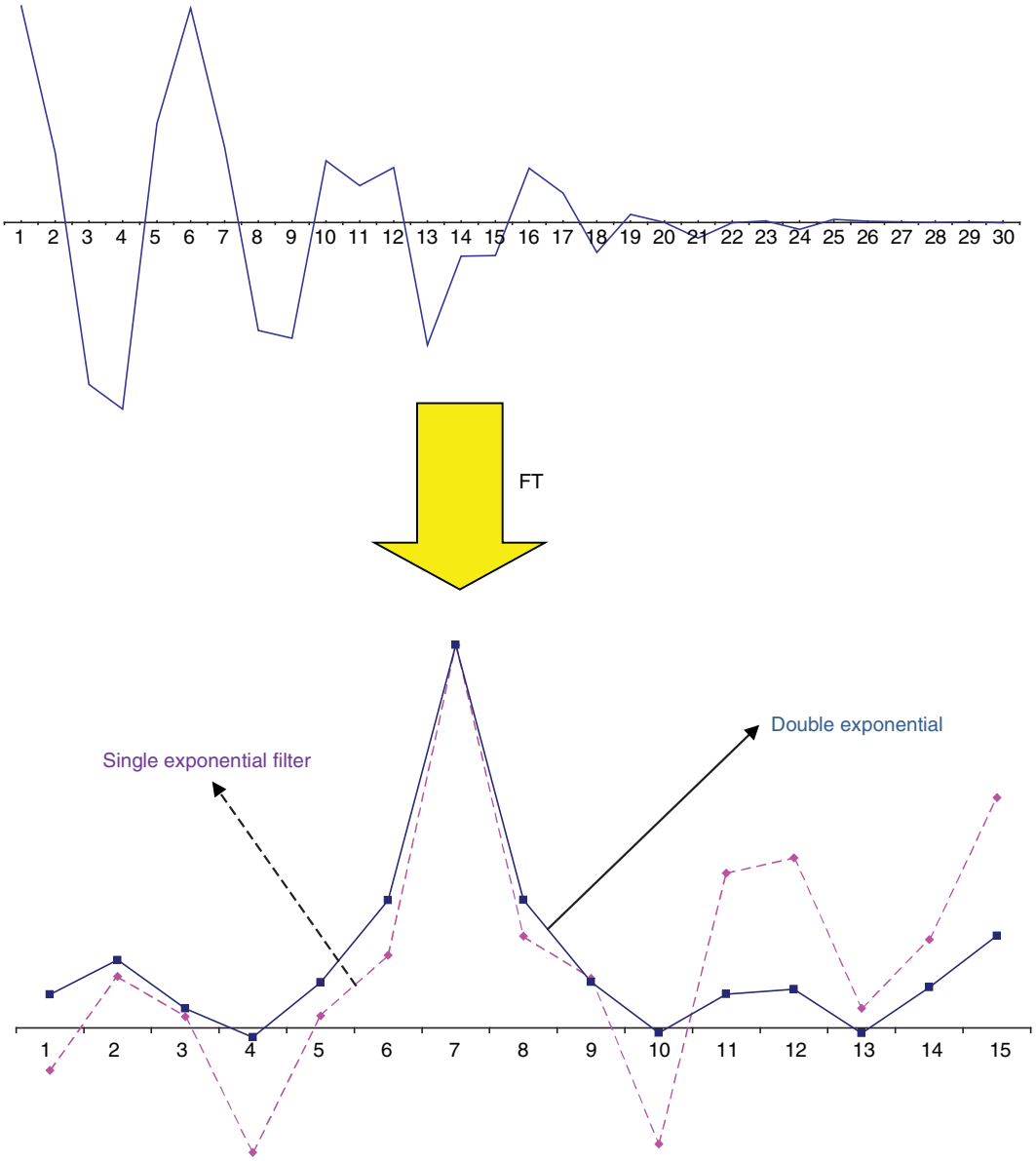


Figure 3.27 Multiplying the data in Figure 3.25 by a double exponential.

1. A typical double exponential filter consists of two exponent terms, the linear one to reduce noise and the quadratic to increase resolution.
   (a) True
   (b) False

3.5.2.3 Fourier Self-Deconvolution

In many forms of spectroscopy such as NMR and IR, data are often acquired directly as a time series and must be Fourier transformed to obtain an interpretable spectrum. However, any spectrum or chromatogram can be processed using Fourier filters, even if not acquired as a time series. The secret is to inverse transform (see Section 3.5.1.2) back to a time series.

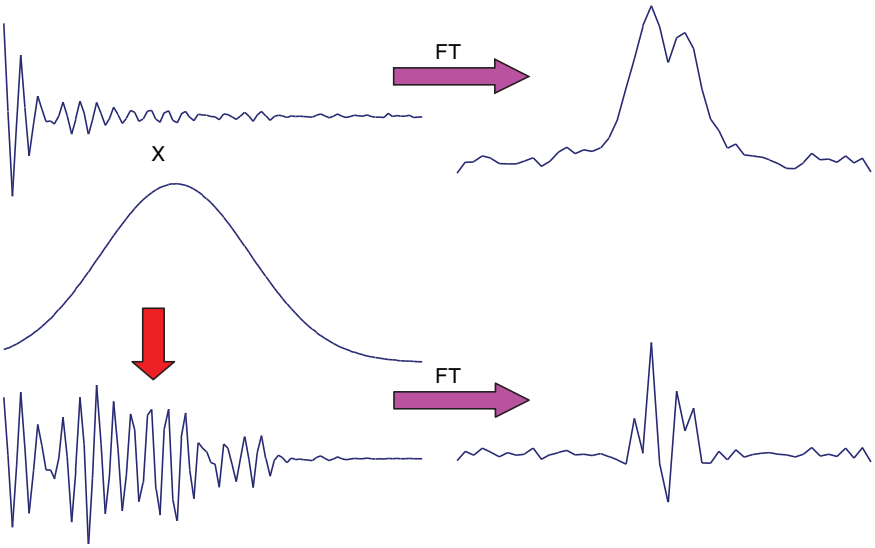


Figure 3.28 Use of a double exponential filter.

Normally, three steps are employed, as illustrated in Figure 3.29.

- Transform the spectrum into a time series. This time series does not physically exist but can be handled by a computer.
- Then apply a Fourier filter to the time series.
- Finally transform the spectrum back, resulting in improved quality.

This procedure is called *Fourier self-deconvolution* and is an alternative to digital filters of Section 3.3.

1. Any frequency domain signal can be improved in shape using Fourier filters.
   (a) True
   (b) False

3.5.3 Convolution Theorem

Some people get confused by the difference between Fourier filters, linear smoothing and resolution functions. In fact, both methods are equivalent and are related by the convolution theorem, and both have similar aims to improve the quality of spectroscopic or chromatographic or time series data.

The principles of convolution have been discussion in Section 3.3.3. Two functions, f and g, are said to be convoluted to give h, if

$$h_i = \sum_{j=-p}^{j=p} f_j \, g_{i+j}$$

Convolution involves moving a window or digital filter function (such as a Savitzky–Golay or MA) along a series of data such as a spectrum, multiplying the data by that function at each successive data point. In the applications discussed in this section, window functions will always be applied in the spectral domain rather than time domain. A three-point MA involves multiplying each set of three points in a spectrum by a function containing the values (1/3, 1/3, 1/3), and the spectrum is said to be convoluted by the MA filter function.

Filtering a time series, using Fourier time domain filters, however, involves multiplying the entire time series by a single function, so that

$$H_i = F_i G_i$$

1. In the equation $h_i = \sum_{j=-p}^{j=p} f_j g_{i+j}$ as discussed above.
   (a) f represents the spectral domain and g a window function
   (b) g represents the spectral domain and f a window function
   (c) f represents the time domain and g a window function
   (d) g represents the time domain and f a window function

3.6 Additional Methods

A number of more sophisticated methods have been implemented over the past few decades, although some of the ideas such as Bayes' theorem are very much older. In certain instances, a more specialised approach is appropriate and also generates much interest in the literature. There are particular situations, for example, where data are very noisy or incomplete, or where rapid calculations are required, which require particular solutions. The methods listed below are topical and implemented within a number of common software packages. They do not represent a comprehensive review but are added for completion, as they are regularly reported in the chemometrics literature and are often available in common software packages. The discussion on Bayesian methods is introduced within this section, although can be more widely associated with most statistics: however, unlike, for example, the machine learning or applied statistics community, they are rarely used within chemometrics; thus, we restrict to its application to signal analysis to avoid repetition throughout the text.

3.6.1 Kalman Filters

The Kalman filter has its origin in the need for rapid online curve fitting. In some situations such as chemical kinetics, it is desirable to calculate a model whilst the reaction is taking place rather than wait until the end. In online applications such as process control, it may be useful to see a smoothed curve as the process is taking place, in real time, rather than later. The general philosophy is that, as something evolves with time, more information becomes available so that the model can be refined. As each successive sample is recorded, the model improves. It is possible to predict the response from information provided at previous sample times and see how this differs from the observed response, thus changing the model.

Kalman filters are quite complex to implement computationally, but the principles are as follows and will be illustrated by the case where a single response (y) depends on a single factor (x). There are three main steps.

- Model the current data point (i), for example, calculate $\hat{y}_{i|i-1} = \boldsymbol{x}_i \boldsymbol{b}_{i-1}$ using a polynomial in x, and methods introduced in Chapter 2. The parameters $\boldsymbol{b}_{i-1}$ are initially guesses, which are refined with time. The '|' symbol means that the model of y_i is based on the first $i-1$ data points and $\boldsymbol{x}_i$ is a row vector consisting of the terms in a model (usually but not exclusively polynomial) and $\boldsymbol{b}$ is a column vector. For example, if $x_i = 1.5$ (e.g. the time a sample is measured), then a three-parameter quadratic model of the form $y_i = b_0 + b_1 x + b_2 x_2^2$ gives the vector $\boldsymbol{x}_i = (1, 1.5, 2.25)$.
- This next step is to see how well this model predicts the current data point and calculate $d_i = y_i - \hat{y}_{i|i-1}$, which is called the *innovation*. The closer these values, the better the model.
- Finally, refine the model by recalculating the coefficients

$$\boldsymbol{b}_i = \boldsymbol{b}_{i-1} + \boldsymbol{k}_i d_i$$

If the estimated and observed values of y are identical, the value of $\boldsymbol{b}$ will be unchanged. If the observed value is more than the estimated value, it makes sense to increase the size of the coefficients to compensate. The column vector $\boldsymbol{k}_i$ is called the *gain* vector. There are a number of ways of calculating this, but the larger it is, the greater the uncertainty in the data.

A common (but complicated way) of calculating the gain vector is as follows.

- Start with a matrix $\boldsymbol{V}_{i-1}$, which represents the variance (or error) of the coefficients. This is a square matrix, with the number or rows and columns equal to the number of coefficients in the model. Hence, if there are five coefficients, there will be 25 elements in the matrix. The higher these numbers, the less certain the prediction of the coefficients. Start with a diagonal matrix containing some high numbers.

perform multiple linear regression as discussed in Chapter 2, so that

$$\boldsymbol{y} = \boldsymbol{Xb}$$

or

$$\boldsymbol{b} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$$

using the standard equation for the pseudo-inverse, giving an equation of $\hat{y}_i = 0.512 + 0.068x - 0.009x_2^2$ only very slightly different to the prediction by Kalman filters when $i = 22$. If all the data are available, there is little point using Kalman filter, the method is mainly useful for online predictions; however, they can be useful for smoothing 'on the fly' and with modern computers can be implemented very fast, so that, for example, a smoothed profile is presented on a screen in real time during a process.

Kalman filters can be extended to more complex situations with many variables and many responses. The model does not need to be multi-linear but, for example, may be exponential (e.g. in kinetics). Although the equations increase considerably in complexity, the basic ideas are the same. Although Kalman filters are in fact widely used in engineering, they are no longer very common in the chemometrics literature; thus, we will not discuss the more elaborate enhancements.

1. A process is modelled at point $i-1$ by $\hat{y} = 0.5 + 0.2x - 0.05x^2$. The estimated value at point $i-1$ is 0.8 and the observed value at point i is 0.95. If $\boldsymbol{k}' = (0.03\ 0.06\ -0.04)$, what is the new model at point i?
   (a) $\hat{y} = 0.4955 + 0.191x + 0.01x^2$
   (b) $\hat{y} = 0.5045 + 0.209x - 0.11x^2$
   (c) $\hat{y} = 0.53 + 0.26x - 0.45x^2$
   (d) $\hat{y} = 0.47 + 0.14x + 0.35x^2$
2. A process is modelled by a full third-order polynomial. $\boldsymbol{V}$ has dimensions
   (a) 3×1
   (b) 3×3
   (c) 4×4

3.6.2 Wavelet Transforms

Another topical method in chemical signal processing is the wavelet transform. The general principles are discussed below, without providing detailed information about algorithms, and provide a general understanding of the approach: wavelets are implemented in several chemometric packages; hence, many people have come across them. We will restrict the discussion to the DWT (discrete wavelet transform).

The first transform now classed as a wavelet transform was described in the early twentieth century by Haar and is known as a Haar transform. During the mid-1980s, a collection of related methods were described, together called *wavelet transforms*, of which the Haar was the simplest. As is usual in the machine learning literature, grants, PhDs and conference presentations are obtained by describing increasingly complex algorithms. For complex data such as image processing, for example, in medicine or facial recognition, some of the more elaborate methods provide real benefit, but most chemometrics data are in fact relatively simple. In chemometrics, wavelet transforms are used for two primary purposes, to denoise data and to compress data.

The principle of the Haar transform is in fact very simple. For DWTs, the number of data points, which we will denote as $\boldsymbol{x}$, is a power of 2 and we will restrict discussion to this situation. Consider the data of the left-hand column in Table 3.11. A simple approach of reducing the 16 numbers to 8 would be to average each two successive data points, or in our example we add these (we will discuss scaling below). This provides eight *approximation coefficients*, as in the centre column. By the process, the data have been compressed from 16 to 8 data points, and noise has been reduced by a two-point MA. This represents a level 1 wavelet transform. The data can be smoothed and compressed furthermore to four data points (in our example, this is probably too far) via a level 2 wavelet transform. The results are presented graphically in Figure 3.32. For the level 1 transform, peaks can be clearly distinguished. Obviously, in most practical situations, there will be far greater digital resolution; thus, it may be easier to distinguish peak shapes. However, if we used the level 1 transform, we would be able to distinguish signals better and reduce the amount of data we store.

Normally, to keep the scale of the data constant, the transformed data are multiplied, in this case by $(1/\sqrt{2})^l$ for level l. This makes the sum of squares at each level approximately constant: in our example, it is 5.911 for the raw data, 5.644 for level 1 and 5.264 for level 2. The data transform can be expressed in matrix terms, where the original data $\boldsymbol{x}$ has

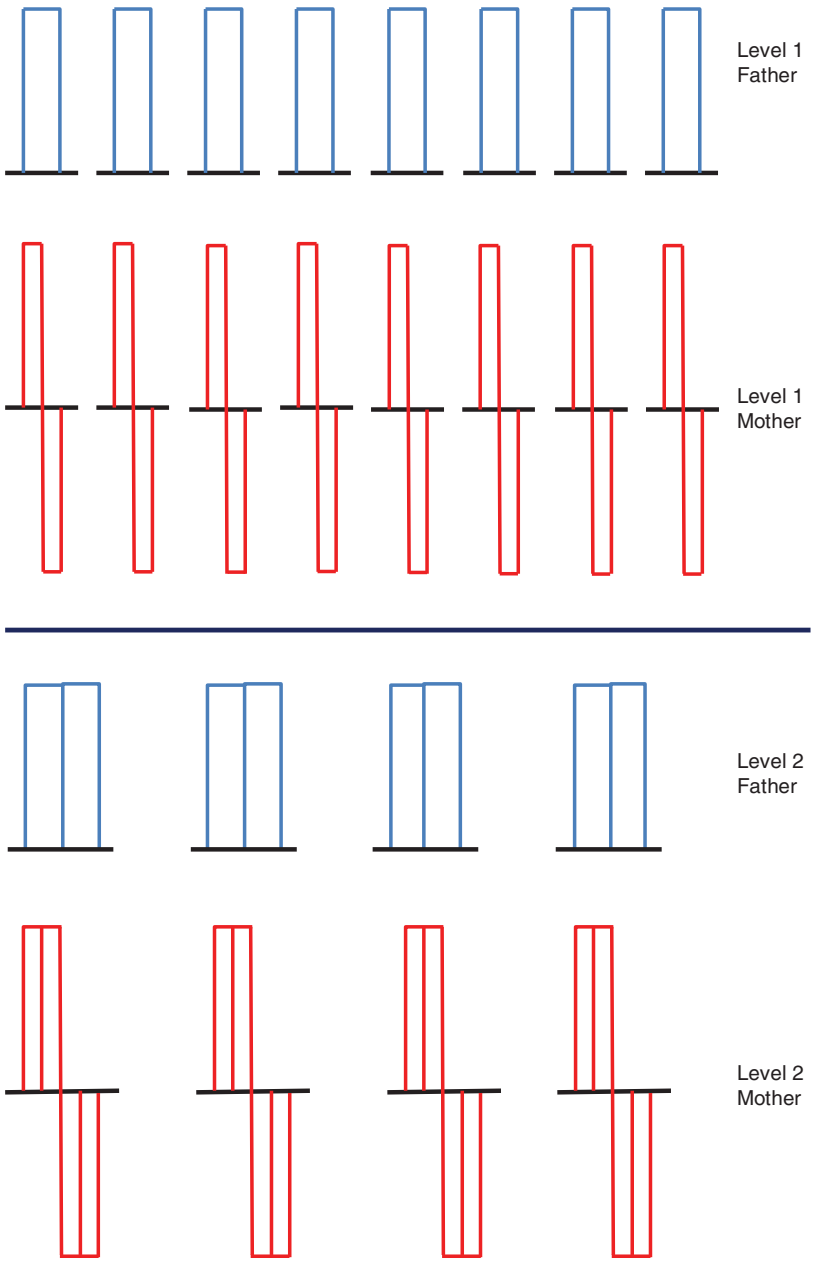


Figure 3.33 Haar wavelets of levels 1 and 2 corresponding to data in Table 3.11.

3.6.3 Bayes' Theorem

Bayesian approaches are now commonplace in most areas of modern statistics and machine learning, but less well used in chemometrics, partly because of the difficulty of incorporating probabilities into algorithms such as partial least squares (Chapter 6), hence the relatively limited discussion in this text. The principle is that before we perform an experiment, we should know something about a system; this prior knowledge can be converted into a probability. The default spectrum might be a flat distribution. As more information is gathered, we build up a more detailed picture of a system, which allows us to predict information usually to greater degrees of certainty. Each observation alters the probability. Many statisticians consider this as a common sense. For example, if we want to predict the temperature at the North Pole in winter, we have prior information that we can use before meteorological modelling, which then improves our predictions and so on. In statistics, these predictions are usually expressed in probabilities; for example, we may want to give an estimate that we are 99% confidence that the temperature on a given day will be within a

The Bayesian method allows prior information or measurements or hypotheses to be taken into account. It also allows continuing experimentation, improving a model all the time, as the Bayesian model can be refined each time more data are obtained. A spectrum can be considered a probability distribution; thus, the aim of experimentation is to refine a prior spectral model.

It is important to emphasise that Bayesian methods are applicable throughout science and are introduced in this section primarily because of their relationship with maximum entropy in signal processing, as discussed below, but should not be viewed as an exclusive tool for signal analysis. This text is primarily about modern chemometric trends, and Bayesian methods are rarely encountered in the chemometrics literature, whilst very widespread in mainstream statistics.

1. 70% of all samples sent to a laboratory are suspected to be adulterated. A spectroscopic test suggests that if a sample is indeed adulterated, this test correctly identifies 80% of adulterated samples, but incorrectly 30% of non-adulterated samples. What is the posterior probability that this sample is actually adulterated?
   (a) 0.727
   (b) 0.903
   (c) 0.861
   (d) 0.823

3.6.4 Maximum Entropy

Over the past decades, there has been substantial scientific interest in the application of maximum entropy techniques with notable successes, for the chemist, in areas such as NMR spectroscopy and crystallography. Maxent has had a long statistical vintage, one of the modern pioneers being Jaynes, but the first significant scientific applications were in the area of deblurring of infrared images of the sky, involving development of the first modern computational algorithm, in the early 1980s. Since then, there has been an explosion of interest and several implementations available within commercial instrumentation. The most spectacular successes have been in the area of image analysis, for example NMR tomography, as well as forensic applications such as obtaining clear car number plates from hazy police photos. In addition, there has been a very solid and large literature in the area of analytical chemistry.

3.6.4.1 Definition

Maxent is one method for determining the probability of a model. A simple example involves the toss of a six-sided unbiased die. What is the most likely underlying frequency distribution, and how can each possible distribution be measured? Figure 3.34 illustrates a flat distribution and Figure 3.35 illustrates a skew distribution (expressed as proportions). The concept of entropy can be introduced in a simple form and is defined by

$$S = -\sum_{i=1}^{I} p_i \log(p_i)$$

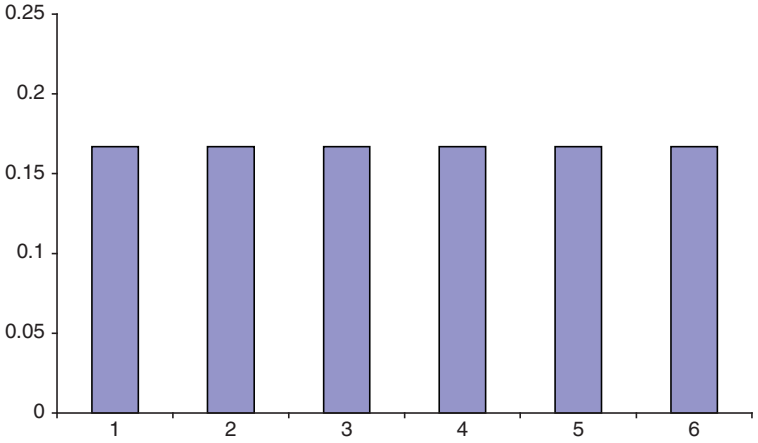


Figure 3.34 Frequency distribution for the toss of a die.

Table 3.13 Maximum entropy calculation for biased die.

	p		*p* log(*p*)	
	A	**B**	**A**	**B**
1	0.00	0.0238	0.000	0.039
2	0.00	0.0809	0.000	0.088
3	0.25	0.1380	0.151	0.119
4	0.25	0.1951	0.151	0.138
5	0.25	0.2522	0.151	0.151
6	0.25	0.3093	0.151	0.158
Entropy			0.602	0.693

This, in fact, is the maximum entropy solution but does not yet take account of experimental data but is the most likely distribution in the absence of more information or is prior probability distribution.

It is important to realise that there are a number of other definitions of entropy in the literature, only the most common being described in this chapter.

1. A six-sided die is tossed 10 times. It comes up 1, 3, 4 and 5 two times, and 2 and 6 one time. What is the entropy of the distribution, using logs to the base 10.
   (a) −0.759
   (b) 0.759
   (c) −2.408
   (d) 2.408
2. Using the maximum entropy criterion, and the hypothesis that the die is unbiased, compare the result in question 1 to a result where the die came up 1, 3, 4, 5 and 6 two times and never 2. Which is the most likely distribution?
   (a) The distribution of question 1
   (b) The distribution of question 2

3.6.4.2 Modelling

In practice, there are an infinite, or at least very large, number of statistically identical models that can be obtained from a system. If I know a chromatographic peak consists of two components, I can come up any number of ways of fitting the chromatogram, all with identical least squares fit to the data. In the absence of further information, a smoother solution is preferable and most definitions of entropy will pick such an answer.

Although in the absence of any information at all, a flat spectrum or chromatogram is the best answer, experimentation will change the solutions considerably, and should pick two underlying peaks that fit the data well consistent with maximum entropy. Into the entropic model, information can be built relating to knowledge of the system. Normally, a parameter is calculated as

$$\text{Entropy function} - \text{statistical fit function}$$

High entropy is good, but not at the cost of a numerically poor fit to the data; however, a model that fits the original (and possibly very noisy) experiment well (e.g. a polynomial with low least square errors) is not a good model if the entropy is too high. The statistical fit can involve a least squares function such as χ^2, which is hoped to minimise. In practice, for a number of models with identical fit to the data, the one with maximum entropy is the most likely. Maximum entropy is used to calculate a *prior* probability (see discussion on Bayes' theorem) and experimentation refines this to give a *posterior* probability. Of course, it is possible to refine the model still further by performing yet more experiments, using the posterior probabilities of the first set of experiments as prior probabilities for the next experiments. In reality, this is what many scientists do, continuing experimentation until they reach a desired level of confidence, the Bayesian method simply refining the solutions.

For relatively sophisticated applications, it is necessary to implement the method as a computational algorithm, being a number of packages available in instrumental software. The algorithms are quite complex and out of the scope of this